
Tutorial No.: 9
Engineering Mechanics (ME-101)
Date: 26/03/2024 and Time: 7.55 to 8.55 AM

Question 1: Each of the three balls in Figure 1, has mass m and is welded to the rigid equiangular frame of negligible mass. The assembly rests on a smooth horizontal surface. If a force F is suddenly applied to one bar shown, determine (a) the acceleration of the point O and (b) the angular acceleration $\ddot{\theta}$ of the frame. [10 marks]

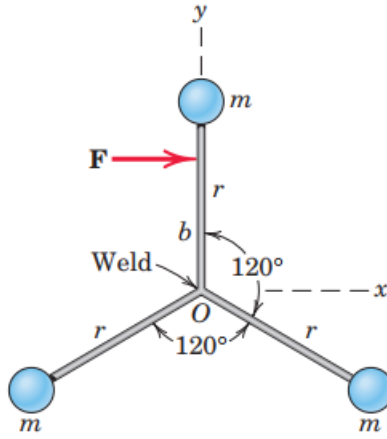


Figure 1

Solution 1: Note, O represents the center of mass of the system. Writing equation of motion in the x direction,

$$(a) \sum F_x = (\sum m_i) a_{Ox} \Rightarrow a_{Ox} = \frac{F}{3m} \quad (\text{along positive } x \text{ direction}) \quad [5 \text{ marks}]$$

Since, $\sum F_y = 0$, $a_{Oy} = 0$. Now, considering moment of all external forces about center of mass O ,

$$(b) \sum M_c = \dot{H}_c \Rightarrow bF = 3 \times \frac{d}{dt}(r \times mr\dot{\theta}) \Rightarrow \ddot{\theta} = \frac{bF}{3mr^2} \quad (\text{clockwise}) \quad [5 \text{ marks}]$$

Question 2: The small car shown in Figure 2, has a mass of 20 kg, rolls freely on the horizontal track and carries the 5-kg sphere mounted on the light rotating rod with $r = 0.4$ m. A geared motor drive maintains a constant angular speed $\dot{\theta} = 4$ rad/s of the rod. If the car has a velocity $v = 0.6$ m/s when $\theta = 0^\circ$, calculate v when $\theta = 60^\circ$. Neglect the mass of the wheels and any friction. [10 marks]

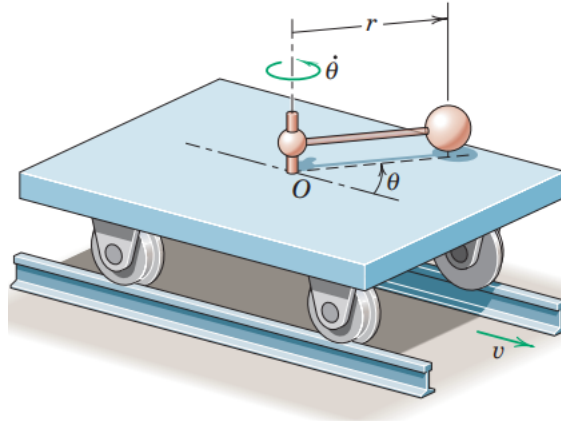
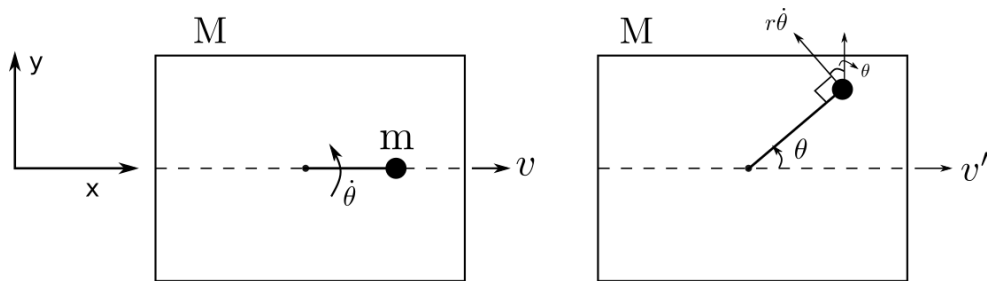


Figure 2

Solution 2:



Since there are no external forces on the cart and the ball assembly, thus its linear momentum in the x-direction would be conserved.

$$G_{2x} = G_{1x}$$

$$Mv' + m(v' - r\dot{\theta} \sin \theta) = (M + m)v \quad [6 \text{ marks}]$$

$$v' = \frac{(M + m)v + mr\dot{\theta} \sin \theta}{M + m} = 0.877 \text{ m/s} \quad [4 \text{ marks}]$$

Can we apply linear momentum conservation along y-direction?

⇒ **No**, because of the reaction forces on the wheel in the y-direction.

Question 3: The two spheres, depicted in Figure 3, are rigidly connected to the rod of negligible mass and are initially at rest on the smooth horizontal surface. A force F is suddenly applied to one sphere in the y -direction and imparts an impulse of 10 N-s during a negligibly short period of time. As the spheres pass the dashed position, calculate the velocity of each one.

[10 marks]

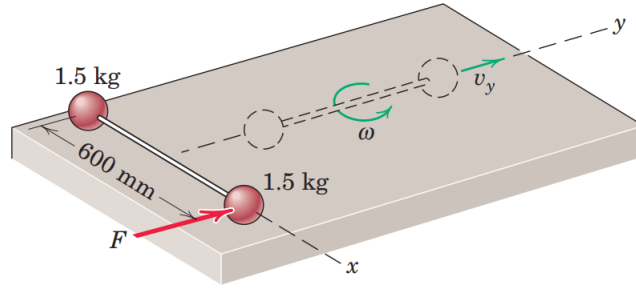
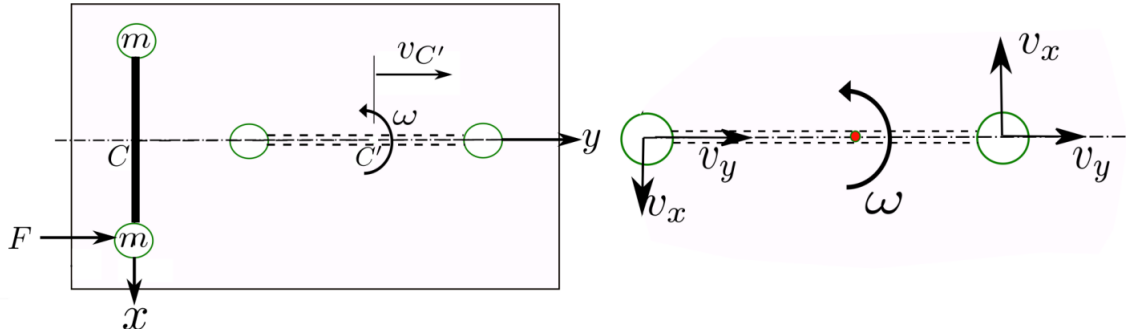


Figure 3

Solution 3:



Using linear-impulse-momentum relation

$$\int \Sigma F_y dt = G_{2y} - G_{1y}, \quad (1)$$

$$\Rightarrow 10 = 2(mv_{C'}) - 0, \quad [2 \text{ marks}]$$

$$\Rightarrow v_{C'} = \frac{5}{1.5} = \frac{10}{3} \text{ m/s}. \quad [2 \text{ marks}] \quad (2)$$

Invoking angular impulse-angular momentum relation about centre of mass,

$$\int \Sigma M_C dt = H_{C'} - H_C.$$

$$\Rightarrow \int F \left(\frac{l}{2} \right) dt = 2 \times \underbrace{\left(\frac{l}{2} \times mv_x \right)}_{C.C.W} - 0. \quad [2 \text{ marks}]$$

$$\Rightarrow v_x = \frac{10}{3} \text{ m/s}. \quad [2 \text{ marks}]$$

Note, $v_y = v_{C'}$. Thus, the magnitude of velocity of both the balls is,

$$|\bar{v}_b| = \sqrt{v_x^2 + v_y^2} = \frac{10\sqrt{2}}{3} \text{ m/s} = 4.71 \text{ m/s}. \quad [2 \text{ marks}]$$

Note: Angular velocity of the assembly can be obtained as $\omega = \frac{v_x}{l/2} = \frac{100}{3} \text{ rad/s}$.

Question 4: The three small spheres, each of mass m , are secured to the light rods to form a rigid unit supported in the vertical plane by the smooth circular surface (refer Figure 4). The force of constant magnitude P is applied perpendicular to one rod at its midpoint. If the unit starts from rest at $\theta = 0$, determine (a) the minimum force P_{min} which will bring the unit to rest at $\theta = 60^\circ$, and (b) the common velocity v of spheres 1 and 2, when $\theta = 60^\circ$, if $P = 2P_{min}$.

[10 marks]

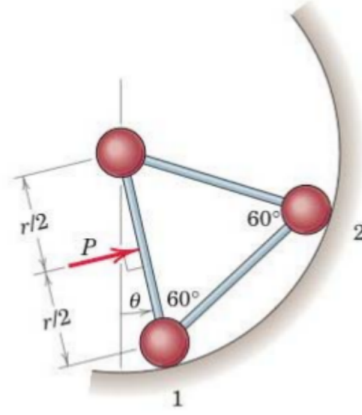
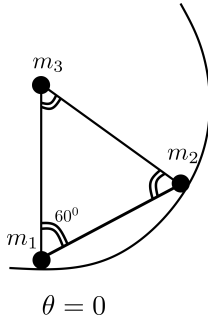


Figure 4

Solution 4:

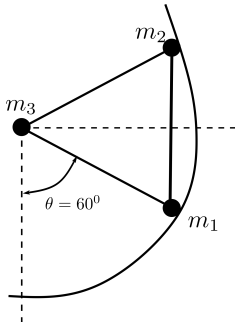


$$(a) \quad W_{1-2} = V_2 - V_1 = \Delta V$$

$$\Rightarrow P_{min} \left(\frac{r}{2} \times \frac{\pi}{3} \right) = m_1 g (r - r \cos \frac{\pi}{3}) + m_2 g r \quad [3 \text{ marks}]$$

$$= \frac{3}{2} mgr \quad \dots (m_1 = m_2 = m)$$

$$P_{min} = \frac{3}{\pi r} mgr \frac{3}{2} = \frac{9mg}{\pi} \quad [2 \text{ marks}]$$



$$(b) \quad 2P_{min} \left(\frac{r}{2} \times \frac{\pi}{3} \right) = \Delta V + \Delta T \quad [2 \text{ marks}]$$

$$\text{now, } \Delta V = P_{min} \left(\frac{r}{2} \times \frac{\pi}{3} \right)$$

$$\Rightarrow \Delta T = P_{min} \left(\frac{r}{2} \times \frac{\pi}{3} \right)$$

$$2 \times \frac{1}{2} mv^2 = \frac{9mg}{\pi} \times \frac{\pi r}{6} \quad [2 \text{ marks}]$$

$$\Rightarrow v = \sqrt{\frac{3gr}{2}} \quad [1 \text{ mark}]$$

Question 5: As shown in Figure 5, a high speed jet of air issues from the nozzle A with a velocity of v_A and mass flow rate of 0.36 kg/s . The air impinges on a vane causing it to rotate to the position shown. The vane has a mass of 6 kg . Knowing that the magnitude of the air velocity is equal at A and B , determine, (a) the magnitude of the velocity at A , and (b) the components of the reactions at O . [10 marks]

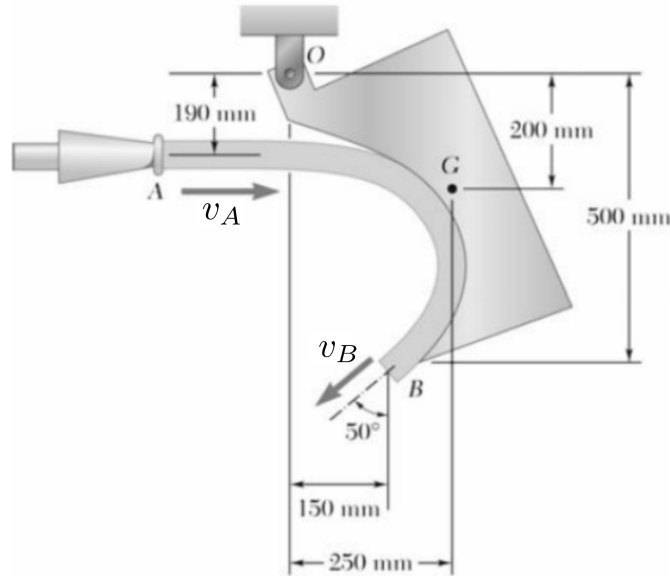
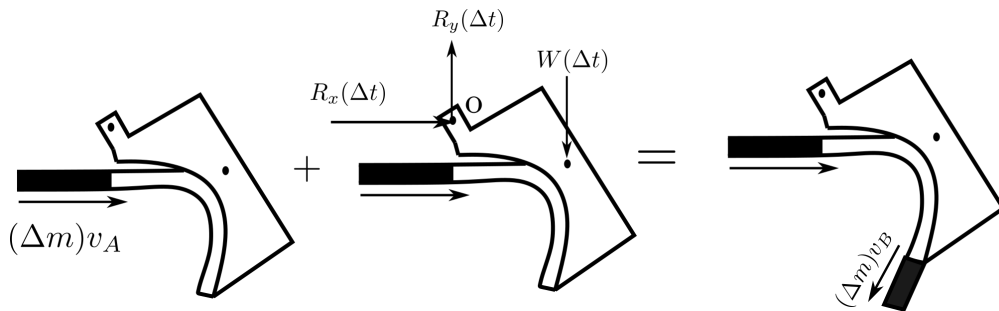


Figure 5

Solution 5: Assume that the speed of the air jet is same at A and B . i.e, $v_A = v_B = v$
Apply the principle of impulse and momentum.



(a) Using angular-impulse-momentum relation about O (+ $\curvearrowright$):

$$(0.190)(\Delta m)v - (0.250)W(\Delta t) = -(0.150)(\Delta m)v \cos 50^\circ - (0.500)(\Delta m)v \sin 50^\circ \quad [2 \text{ marks}]$$

$$\begin{aligned} v &= \frac{\Delta t}{\Delta m} \cdot \frac{0.250W}{0.150 \cos 50^\circ + 0.500 \sin 50^\circ + 0.190} \\ &= \frac{0.250W}{0.669} \left(\frac{1}{\dot{m}} \right) \\ &= \frac{(0.250)(6)(9.81)}{(0.669)(0.36)} \Rightarrow v = 61.058 \text{ m/s} \end{aligned}$$

[1 mark]

(b) Using linear-impulse-momentum relation along x :

$$(\Delta m)v + R_x(\Delta t) = -(\Delta m)v \sin 50^\circ \quad [2 \text{ marks}]$$

$$R_x = -\frac{\Delta m}{\Delta t}v(1 + \sin 50^\circ)$$

$$= -(0.36)(61.058)(1 + \sin 50^\circ)$$

$$\Rightarrow R_x = -38.82 \text{ N} \quad [1 \text{ mark}]$$

(c) Using linear-impulse-momentum relation along y :

$$0 + R_y(\Delta t) - W(\Delta t) = -(\Delta m)v \cos 50^\circ \quad [2 \text{ marks}]$$

$$R_y = W - \frac{\Delta m}{\Delta t}v \cos 50^\circ$$

$$= (6)(9.81) - (0.36)(61.058) \cos 50^\circ$$

$$\Rightarrow R_y = 44.73 \quad [1 \text{ mark}]$$

$$R = \sqrt{(38.82)^2 + (44.73)^2} = 59.2 \text{ N}$$

$$\tan \alpha = \frac{44.73}{38.82} \Rightarrow \alpha = 49^\circ \quad [1 \text{ mark}]$$

Question 6: The end of a chain of length L and mass ρ per unit length which is piled on a platform is lifted vertically with a constant velocity v by a variable force P (refer Figure 6). Find P as a function of the height x of the end above the platform. Also find the energy lost during the lifting of the chain. [10 marks]

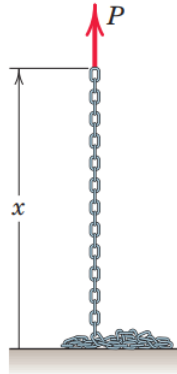


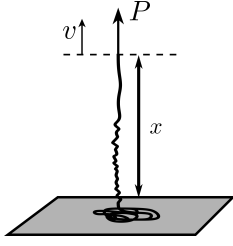
Figure 6

Solution 6:

In case of variable mass system we have,

$$\sum F = m \frac{dv}{dt} + \dot{m}u \quad \dots\dots(v = \text{constant}) \quad (3)$$

[1 mark]



Here, the rate of mass intake, $\dot{m} = \rho \dot{x} = \rho v$

The magnitude of the relative velocity of the mass stream,

$$u = v - v_0 = v$$

Now, using Equation 3

$$\begin{aligned} \sum F &= P - \rho x g = (\rho v)v \\ \implies P &= \rho(xg + v^2) \end{aligned}$$

[4 marks]

The work done in lifting the whole chain is,

$$W_{1-2} = \int_0^L P dx = \frac{1}{2} \rho g L^2 + \rho v^2 L$$

[2 marks]

Total change in energy of the system,

$$\Delta T + \Delta V = \left[\frac{1}{2} (\rho L) v^2 - 0 \right] + \left[(\rho L) g \left(\frac{L}{2} \right) - 0 \right]$$

[2 marks]

Hence, the total energy loss in lifting the whole chain,

$$\Delta E = W_{1-2} - (\Delta T + \Delta V) = \frac{1}{2} \rho L v^2$$

[1 mark]